Test-Time Scaling for Visual Chain-of-Thought Models

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Links between Computer Vision and Language

MODELS: SAME 3B BACKBONE, DIFFERENT REASONING

Qwen2.5-VL-3B
Direct answer. No intermediate reasoning.
image + Q → ans

GRIT-3B
Grounded visual chain-of-thought.
think → bbox → rethink → ans

HOW GRIT REASONS: WORKED EXAMPLE

Q: "What is the position of the cat in relation to the vase?" (A) behind (B) on top (C) inside (D) in front

input image

1
<think>
The cat appears to be resting inside or on top of the vase opening. I need to look more closely at the region around the cat and vase to disambiguate "on top" from "inside".
</think>

COMPUTE & EMIT BBOX [108, 5, 236, 178] → crop & re-examine

grounded crop

2
<rethink>
Looking at the crop, the cat is inside the lid opening, not in front or behind.
answer: B ✓

BENCHMARK: HARD, RECENT, VISION-INDISPENSABLE

MMMU-Pro · VQA | OCRBench v2 · OCR | MMStar · Counting

TTS RECIPE: 9 CANDIDATES PER QUESTION

Two diversity axes: temperature + image augmentation

- Original · T = 0 (greedy)
- Paraphrase A · T = 0.7
- Brightness · T = 0.7
- Edge enhance · T = 0.7
- **Rotation 90°** · T = 0.7
- Original · T = 0.7
- Paraphrase B · T = 0.7
- Grayscale · T = 0.7
- JPEG recompress · T = 0.7

Metric: **pass@9**. Correct if *any* of the 9 candidates is correct. Upper bound on what a perfect selector could recover.

HEADLINE: TTS LIFTS ACCURACY SHARPLY

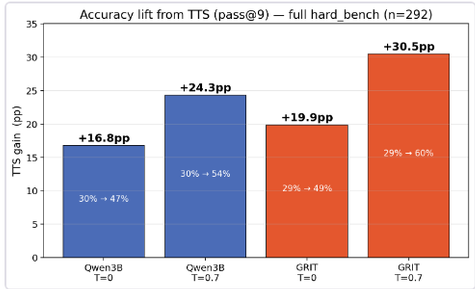

Fig. A — TTS gain (pass@9 − greedy) across all 4 configurations on the full 300-question benchmark. Standard recipe (T = 0.7 + augmentation) reaches **+30.5 pp for GRIT** and **+24.3 pp for Qwen3B**.

FINDING 1
Visual CoT benefits from TTS via image-input diversity; direct models don't.

At T = 0, Qwen produces the *same* answer regardless of augmentation. GRIT still gains **+19.9 pp** from those same augmentations: its grounding step amplifies small visual perturbations into different reasoning chains.

PER-TASK BREAKDOWN: GAINS LARGEST WHERE MODELS HAVE HEADROOM

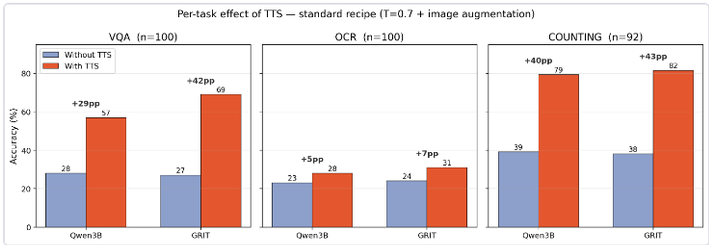

Fig. B — Per-task effect of TTS under the standard recipe. **VQA and Counting**, the two tasks where models have real headroom, show the largest gains (**+29 to +43 pp**). OCR gains stay modest (+5–7 pp) because exact-match scoring penalises augmentations that produce different-but-still-correct surface forms.

VOTING CANNOT CLOSE THE ORACLE GAP

Plurality vote	Logprob sum	Greedy tiebreak	Consistency filter
−1.0 pp	−3.1 pp	−1.7 pp	−1.0 pp

Every answer-level strategy lands at or below greedy. The pool holds the signal, but simple aggregation cannot extract it; a learned verifier is needed.

TEMPERATURE & AUGMENTATION COMPOUND

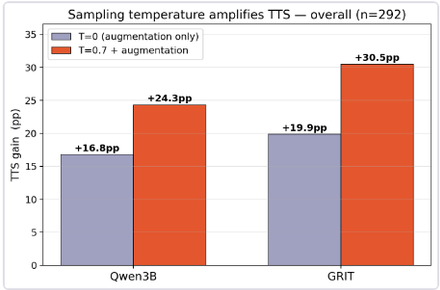

Fig. C — Temperature sampling roughly **doubles** the TTS gain for both models (Qwen +16.8 → +24.3; GRIT +19.9 → +30.5 pp). Under pass@9 both diversity sources compound.

FINDING 2
Structural diversity beats random diversity.

A calibrated entropy study found GRIT is *less* stochastic than Qwen on all tasks (ΔH = −0.23 to −0.47 bits), yet it benefits *more* from TTS. **What matters is not how much diversity a model produces, but whether it is routed through a reasoning surface.**

WHICH AUGMENTATION DRIVES THE GAIN?

Augmentation (T = 0)	GRIT uniq.	Qwen uniq.
Rotation 90°	10	4
JPEG recompress	5	3
Edge × Paraphrase	4	1
Brightness / contrast	3	4
Edge enhance	1	0
Paraphrase (text)	1	1
Grayscale	0	3

Unique rescue = questions where this augmentation is the **only** correct candidate. Rotation alters *spatial layout*; pixel-level augs do not. GRIT's grounding step amplifies spatial change into a new reasoning path.

TAKE-HOME

TTS for visual reasoning works through **image-input diversity channelled by a grounding step**, not through decoding randomness. Expose an intermediate *visual* reasoning surface, and input perturbations become genuinely different reasoning trajectories.

★ GRIT's augmented candidates rescue **different** failures (complementary diversity); Qwen's rescue **the same** failures (redundant). Grounded reasoning turns spatial perturbations into structurally distinct chains; that is what makes TTS work.

LIMITATIONS

→ pass@9 is an oracle upper bound; no selector matches it yet
→ 3B models only; OCR uses strict exact-match scoring

NEXT STEPS

→ Confidence-based / learned verifier for selector design
→ DeepEyesV2 (agentic CoT) full evaluation

DATASETS · MODELS · COMPUTE

MMMU-Pro | OCRBench v2 | MMStar | Qwen2.5-VL-3B | GRIT-3B
Rorqual H100 · DRAC

SCAN FOR CODE